HOW IMPLICIT ASSOCIATIONS OF SOCIAL NORMS CAN BE OPERATIONALIZED IN ENGINEERING DESIGN RESEARCH & EDUCATION

A how-to zine on building your own Implicit Association Test to raise critical consciousness in engineering design

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Knowledge production is a matter of justice. What we know (or think we know or claim to know) shapes the worlds we make. So if we want to change the world, we need to know differently.

Professor Aimi Hamraie

Demystifying Implicit Bias

The role and function of bias

We aren't born believing men should go to work while women stay home. Early in our life, our brains start picking up patterns in the world around us through culture, media, education, and everyday interactions--even if we're not fully aware of them.

This is where **implicit bias** comes in. Implicit bias is your brain's way of simplifying the overwhelming amount of information we process every day. From an **evolutionary perspective**, bias helped us survive, allowing us to quickly determine whether the bear was friend or foe. These mental shortcuts happen automatically, often outside our conscious awareness, and they can shape how we think, feel, and act. To have implicit biases is to be human.

While these shortcuts once helped us react quickly to danger, today in our modern world, those mental shortcuts can lead us to reinforce unfair assumptions or harmful social norms such as: who we see as a "real" engineer or what makes a design "innovative" or "universal."

So, What *is* Implicit Bias?
Specifically, implicit bias refers to attitudes or stereotypes that affect our understanding, actions, and decisions in an unconscious way, making them difficult to control (Greenwald & Banaji, 1995).
→ Bias is a **systematic tendency to favor one thing over another,** a deviation from the mean; can be caused by inadequate data collection and measurement, omission of too many variables or flawed study design.

What Implicit Bias is Not. Implicit bias is **not** racism, sexism, transphobia, etc. These are forms of prejudice, or **explicit and conscious aversion** towards a group of people.

Little Things Matter: How Implicit Bias Shapes Engineering Design

While implicit bias may be innate, when it is not considered and taken into account, it can heavily impact design outcomes over time. Here are just a few examples of modern tech and innovations that have been examined for their bias and impact:

1. **Implicit gender and racial biases in Generative AI chat bots**

ChatGPT

FEBRUARY 7, 2024 | 9 MIN READ

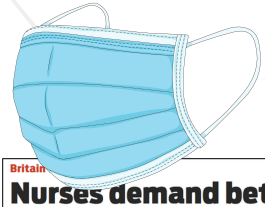
Even ChatGPT Says ChatGPT Is Racially Biased

When asked, ChatGPT declared that its training material—the language w humans use every day—was to blame for potential bias in stories it generat

Gender biases within Artificial Intelligence and ChatGPT: Evidence, Sources of Biases and Solutions

Jerlyn Q.H. Ho [a], Andree Hartanto [a], Andrew Koh [b], Nadyanna M. Majeed [c]

2. **Personal Protective Equipment (PPE) that doesn't fit because it's designed after men**



Sexism on the Covid-19 frontline: 'PPE is made for a 6ft 3in rugby player'

Health professionals, experts and unions say poorly fitting equipment is risking lives of female workers

Britain

Nurses demand better pay and improved PPE

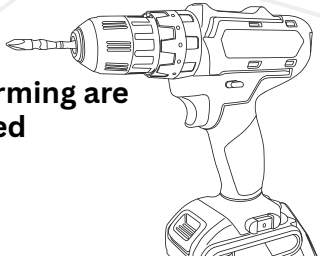
3. **Gendered and ableist public toilet/ bathroom design**

Research Synthesis

A Feminist Critical Analysis of Public Toilet Gender: A Systematic Review

Shawna Lewkowitz [1,2] and Jason Gilliland [1,2,3]

4. **Power tools for construction & farming are often male-normed**



(Perez, 2019)

AND MANY MORE...

Why did I make this zine?

This is a **free, how-to guide** on building your own **Implicit Association Test (IAT)**, which is a tool to help uncover implicit or unconscious associations and attitudes and trigger conversations around identity, design, and equity.

A zine (pronounced "zeen") is a self-published, small-batch, DIY, "magazine-like" print that's about anything you want.

This zine exists because I wanted understand the idea of the **gendered technological imagination** (Balsamo, 2011) and how it shapes the things we design. But like most engineers, I had never been exposed to feminist theory or had formal education on how identity, power, or history shaped the tools we create, or why engineering culture is the way it is.

I believe building critical consciousness in engineering design isn't just about making better products. It's about recognizing how systems of oppression shape the technologies we create and how those technologies are used to advance settler-colonial violence and genocide--from Palestine to police surveillance tech at home--racialized and gendered capitalism, and exploitation. We must build critical conscious as a design community so that design becomes a liberatory practice, not just a tool of the master's house (Rossman & Armstrong, 2025; Hamraie, 2017; Bailey, 2021).

I hope this helps you start your own process. Free Palestine.

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Measuring Implicit Bias: The Implicit Association Test (IAT)

Why did I choose the IAT as my test instrument?
The IAT is one of many methods to measure implicit bias. But it's one of the most widely used experimental psychology methods to reveal implicit attitudes that might otherwise be suppressed by social desirability or lack of conscious awareness. The tool has been adapted to understand implicit attitudes about race, gender, disability, weight, and many other social constructs.

How does the IAT work?
The IAT is a set of rapid-fire computer-based trials designed to measure how strongly we associate concepts (e.g., gender) and evaluations (e.g., good, bad) or stereotypes (athletic, clumsy). The idea is that pairing closely related items in our minds is easier (and therefore faster) than pairing incongruent items. The IAT was developed by a team of psychology researchers (Greenwald, Banaji, Nosek et al. 1998) and the tool has been online since 1998 via Project Implicit.

What does my IAT score mean?
The IAT score is based on how long it takes a person, on average, to sort paired concepts in two different parts of the test. If you respond faster when certain ideas are grouped together (e.g., Male + Technical), it suggests a stronger implicit association in your mind. A positive or negative score reflects the direction of that bias, a score near 0 suggests no strong association either way.

Project Implicit IATs

If you've never taken an IAT take one here!

So now it's your turn....
How to Storyboard your IAT

For examples of how to get started, see Harvard's **Project Implicit**. In my study of implicit gender bias in design, I take a **normative** view of gender. Using Gender Value System theory, which highlights how society organizes gendered beings through labor, roles, and traits (Acker 1990, Ehrnberger 2012), I use Male/Female as my target categories and Blue/Pink to represent masculinized and feminized product attributes. See Kang et al. (2025) for why we used colors as labels and why they work for the IAT.

Example: Gender-Design IAT 1

Target concept category labels: Male vs. Female (e.g., white/Black)
Attribute concept category labels: Blue vs. Pink (e.g., Good/Bad, Pleasant/Unpleasant, Data/Ideas, Science/Liberal Arts)

Male	Female	Blue	Pink
Man	Woman	Industrial	Domestic
Husband	Wife	Performance	Aesthetic
Uncle	Aunt	Technical	Colorful
...	...	...	...

<u>Your IAT</u>

Target concept category labels:_____

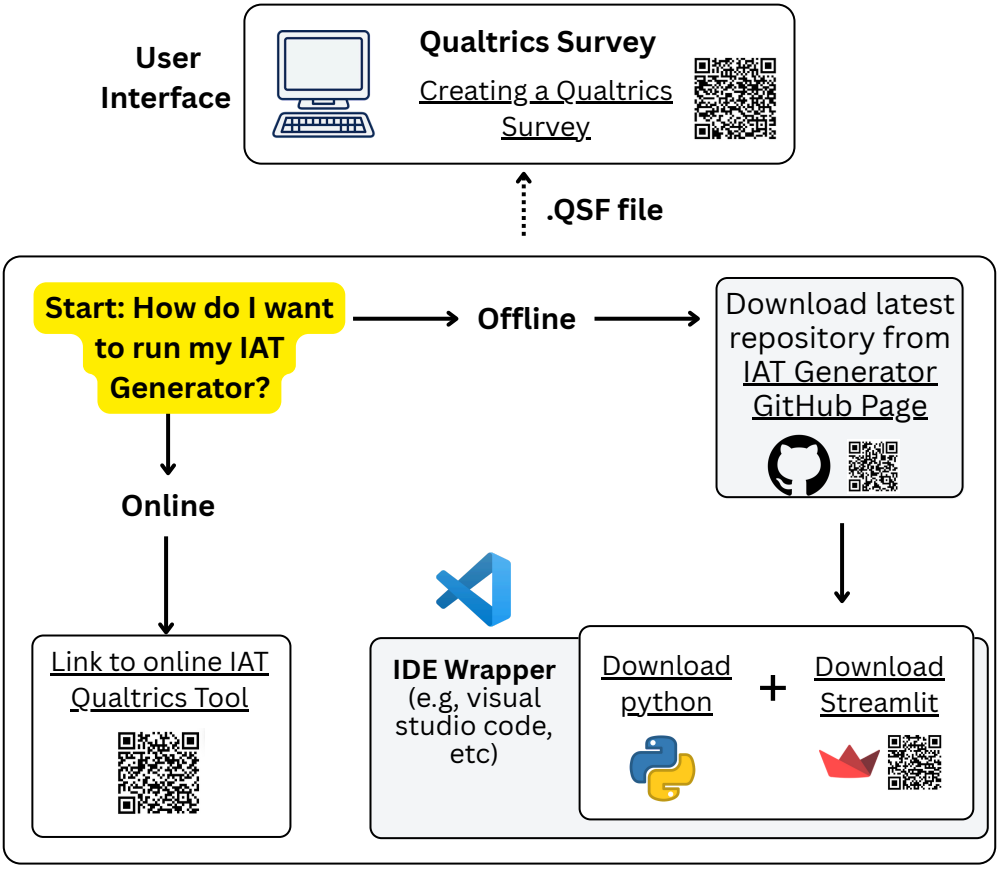
Attribute concept category labels: _____

_____	_____	_____	_____
•	•	•	•
•	•	•	•
•	•	•	•
•	•	•	•

How To Build Your Own Qualtrics-Compatible Implicit Association Test

Welcome to the fun part. To build your own IAT, we utilize an open-source IAT generator created by Cui et al. (2021); all project materials and instructions can be found on their GitHub page. You can run the web app online or host it offline. To host the app offline, you will need to download **python**, an **IDE** like VS code, & **Streamlit**. Obtain access to **Qualtrics** (most institutions have licenses). To analyze the results, you will need a data science program like **R**.

→ **Why host offline?** This option gives you more control and ability to customize your IAT. Also, the online web app crashes sometimes.



Flow chart: How-to Access the IAT Qualtrics Tool so you can create an IAT (.QSF file)

Hosting the IAT Web App Offline

Once you've downloaded python, Streamlit, and Qualtrics:

- Go to the IAT Generator GitHub page and **download** the latest version of the repository (click on **<>Code** drop down > Download ZIP)
- Run the offline web app in a terminal using the command (see Cui's GitHub page for reference):

```
streamlit run your_directory/qualtrics_iat/web_app.py
```

Example (in windows, VS code terminal):

```
C:\Users\kangsa> streamlit run C:
\Users\kangsa\Documents\qualtrics_iat-
master\qualtrics_iat-master\qualtrics_iat\web_app.py
```

*As with all software things, there will be issues in getting things to work, remaining packages you have to download, and directories to be mindful of. Stack Overflow, forums, reddit, etc are super helpful.

Creating Your Own IAT

Once you've launched the web app, fill out the template starting with your target and attribute stimuli as shown in Fig. 1.

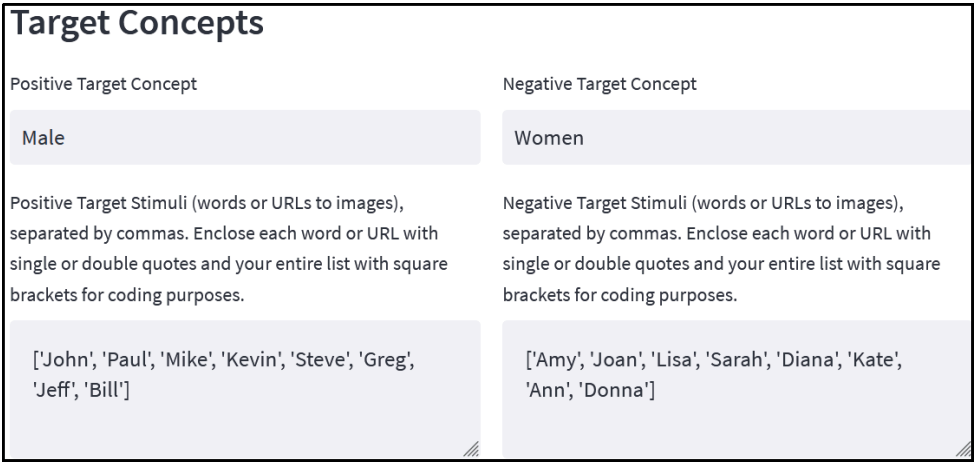


Fig. 1: Screenshot of IAT generator web app for Gender-Career IAT. Greenwald et al (2021) recommends **4-8 easy-to-read, representative stimuli** (words if doing text-based IAT) for your positive and negative concept category.

Creating Your Own IAT cont'd...

Continuing with the Gender-Career IAT example, the researchers used Career & Family as bipolar attribute concepts. The words associated with each positive/ negative concept are shown below.

Project Implicit IATs

Attribute Concepts

Positive Attribute Concept	Negative Attribute Concept
Career	Family
Positive Attribute Stimuli (words or URLs to images), separated by commas. Enclose each word or URL with single or double quotes and your entire list with square brackets for coding purposes.	Negative Attribute Stimuli (words or URLs to images), separated by commas. Enclose each word or URL with single or double quotes and your entire list with square brackets for coding purposes.
['Executive', 'Management', 'Professional', 'Corporation', 'Salary', 'Office', 'Business', 'Career']	['Home', 'Parents', 'Children', 'Family', 'Cousins', 'Marriage', 'Wedding', 'Relatives']

After filling out your Target and Attribute Concept sections out, you will move to **Advanced Configurations**. Most of the default settings already follow IAT best practices, but you can customize:

- Study Name
- Option to switch Attributes or Target stimuli in Block 5
- Counter-Balancing of Blocks, trials per block
- Response Keys (e.g., 'F' key for left hand, 'J' for right hand)
- Instruction and reminder messages to participants
- Attribute and Target label Colors (recommend changing to high contrast colors since background is grey)

Veronica with Four Eyes Blog about Accessible Design for Low-Vision

Contrast Color Finder

Uploading your IAT to Qualtrics

- Once you finished customizing your IAT, click the **Generate Template (.QSF)** button (bottom left) to download your IAT.
- Now in Qualtrics, **create a new survey** and select the option to **import** your own .**QSF file.** Select your desired QSF file.
- **Tip:** Add your front matter to your IAT (consent form, intro) and 'Page Breaks' before and after the test as shown below.

Ex. Gender-Design IAT front matter in Qualtrics

Page Break

Intro

Hello and welcome to the Gender-Design Implicit Association Test (IAT)! In this study you will complete two IATs in which you will be asked to sort words into groups as fast as you can. You will make mistakes and that's OK, just keep going. There are seven parts. The instructions change for each part, so pay attention! Here are the terms that will appear in the test:

Category	Items
Male	Man, Son, Father, Boy, Uncle, Grandpa, Husband, Male
Female	Mother, Wife, Aunt, Woman, Girl, Female, Grandma, Daughter
Blue	Industrial, Innovative, Technical, Competitive, Tough, High-Performance, Powerful
Pink	Stylish, Colorful, Aesthetic, Handcrafted, Domestic, Flexible, Helpful

In addition to the IAT, there are some questions about your experience and some standard demographic questions. Thank you very much for your time! All questions can be sent to **email:** kangsa@oregonstate.edu

Page Break

IAT1-preview

Select a graphic to use for this question

This is the IAT test block

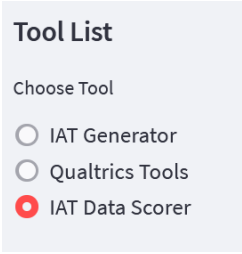
</>

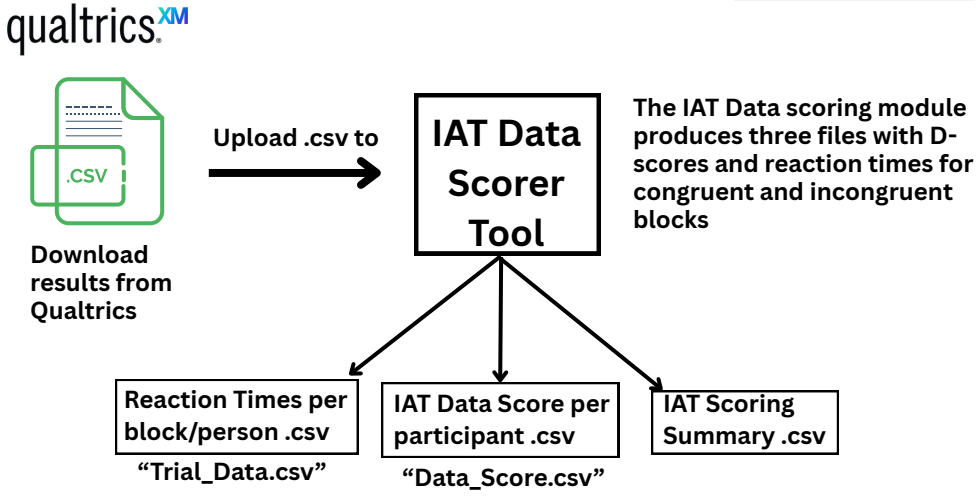
To edit the IAT's javascript (for formatting, etc), **click </>**

Hit **Preview** to view your survey and **Publish** once you're ready!

Using the IAT Data Scorer Tool

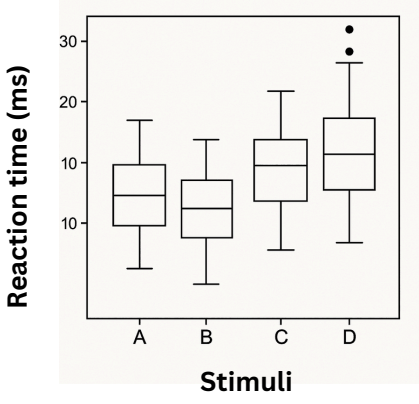
After participants take your Qualtrics survey with your IAT embedded, you can **export** a .csv file of your participants' results from Qualtrics. Upload this .csv file to the **IAT Data Scorer module** in your online or offline-hosted IAT web app.





Using R to Process the IAT Results

Plot your participants' results ("Trial_Data.csv" and "Data_Score.csv") in R using the script on pg. 16 or write your own! Your boxplots show which stimuli (words) participants had a harder time pairing with your target label and your population D-score stats.



Facilitating Discussion around the IAT

It's important to provide a supportive and non-judgemental space for students to reflect on their experience taking the IAT, their results, and ways to move forward (Gill et al. 2010). The goal is to also decouple moral orientation from implicit bias and provide actionable steps forward. There are many ways to go about this, but some sample questions you could ask.

Before the IAT, you might ask your students:

- How do you think your implicit associations of social norms (i.e., gender) might influence your design decisions, if at all?

After the IAT, you might ask:

- How did you feel while taking the IAT? (neutral, discomfort, curious, surprised, overwhelmed, etc)

Other discussion Qs:

- Something that surprised you: ?
- Something this made you think about: ?

Future Action:

This zine is just a starting point. Here are just a few ways to continue the work.

- Engage with the literature on implicit bias in design
- Examine the products and technologies around you
  - Practice noticing the subtle ways social norms shape design (i.e., gendered, racial, and ableist ways)
- Try adapting or designing your own IAT!
  - Use it to study bias in your classroom, lab, or community
  - Use it as a tool to trigger reflection/conversation around implicit bias in design
- Find others doing this work! I also suggest finding a psych/cognitive science researcher to work with :)

Glossary

- **Bias** = a systematic inclination to favor one thing over another. (i.e., implicit gender bias is an unintended systematic neglect of either men or women, Hamberg, 2008)
- **Gender** (as a social construct) = is not fixed nor biologically determined, it's something that's performed and displayed (Butler 1999)
- **Gender (a normative framing)** = gender is fixed, binary (male/female), male-normed and -centric, the "default" user (Perez, 2019)
- **Identity-protective cognition** = engineers who are immersed in a culture of objectivity often pride themselves on only looking at facts, and can have strong emotional reactions and dismiss scientific studies of implicit bias when those studies undermine a shared cultural belief (Walker et al 2017)
- **Implicit Association Test (IAT)** = a widely used experimental psychology tool used to measure the strengths of implicit associations (Greenwald et al., 1998)
- **Social norm** = a shared acceptable behavior by a group. Social norms can both be informal understandings that govern the behavior of members of a society, as well as be codified into rules and laws
- **Stereotype** = a simplified or standardized conception or image invested with special meaning and held in common by members of a group. The tendency for stereotype-confirming thoughts to pass spontaneously through a person's mind is implicit bias.
- **Stereotype threat** = often defined as the condition for being at risk of confirming a negative stereotype about a group to which one belongs. For example, a stereotype threat exists for girls taking a math test because of the stereotype that girls cannot "do math" (Dalal et al 2021).

R script for plotting Trial_Data.csv

Author: Andy Dong & Samantha Kang

```
library(readr)
library(readxl)
library(tidyverse)
library(reshape2)

# Set your working directory
setwd("C:/YourWorkingDirectory")

YourIAT_Trial_Data <- read_csv("YourTrial_Data.csv",
                               col_types = cols(block_number = col_integer(),
                                                trial_number = col_integer(), reaction_time = col_integer(),
                                                task_block_counter = col_integer()))

block_names <- c(
  `3` = "Congruent",
  `6` = "Incongruent")

# standard plot

ggplot(data = YourIAT_Trial_Data %>% filter(block_number ==3 | block_number==6),
       mapping = aes(x=trial_stimulus, y=reaction_time)) +
  geom_boxplot() +
  labs(title="Design IAT Reaction Times",x='Stimulus', y='Reaction Time (ms)') + facet_wrap(~block_number) +
  theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust=1))

# outliers removed

ggplot(data = YourIAT_Trial_Data %>% filter(block_number ==3 | block_number==6) %>%
         filter(!reaction_time %in% boxplot.stats(reaction_time)$out),
       mapping = aes(x=trial_stimulus, y=reaction_time)) +
  geom_boxplot() +
  labs(title="Design IAT Reaction Times",x='Stimulus', y='Reaction Time (ms)') + facet_wrap(~block_number, labeller =
as_labeller(block_names)) +
  theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust=1))

# subtract reaction times between block 6 (incongruent) and block 3 (congruent)

block_3 <- YourIAT_Trial_Data %>% filter(block_number ==3)
block_6 <- YourIAT_Trial_Data  %>% filter(block_number ==6)
dissonance <- merge(block_3,block_6,by.x = c("ResponseId","study","trial_stimulus"), by.y = c("ResponseId","study","trial_stimulus")) %>%
  mutate(diff_reaction_time = reaction_time.y - reaction_time.x)

# outliers removed
ggplot(data = dissonance %>% filter(!diff_reaction_time %in% boxplot.stats(dissonance$diff_reaction_time)$out) %>%
         filter(study=='iat1'),
       mapping = aes(x=trial_stimulus, y=diff_reaction_time)) +
  geom_boxplot() +
  labs(title="Difference in Design IAT Reaction Times Given Product Characteristics",x='Stimulus', y='Reaction Time (ms)') +
  theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust=1)) +
  stat_summary(aes(y = ifelse(diff_reaction_time > (1500*0.9),   # if median in top10% of plot window
                       (diff_reaction_time - (1500/10)),  # put it below bar
                       (diff_reaction_time + (1500/10))), # else put it above
               label = round(..y.., 2)), #round(median(value), 2))
           fun = median, color="red",
           geom = "text")

ggplot(data = dissonance %>% filter(!diff_reaction_time %in% boxplot.stats(dissonance$diff_reaction_time)$out) %>%
         filter(study=='iat2'),
       mapping = aes(x=trial_stimulus, y=diff_reaction_time)) +
  geom_boxplot() +
  labs(title="Difference in Design IAT Reaction Times Given Product Names",x='Stimulus', y='Reaction Time (ms)') +
  theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust=1)) +
  stat_summary(aes(y = ifelse(diff_reaction_time > (1500*0.9),   # if median in top10% of plot window
                       (diff_reaction_time - (1500/10)),  # put it below bar
                       (diff_reaction_time + (1500/10))), # else put it above
               label = round(..y.., 2)), #round(median(value), 2))
           fun = median, color="red",
           geom = "text")

# find correlation between mean values of response times for attributes across IAT1 and IAT2
# Masculine attributes: Industrial, Innovative, Technical, Competitive, Tough, High-Performance, Powerful
# Feminine attributes: Domestic, Aesthetic, Stylish, Helpful, Flexible, Colorful, Handcrafted

masculine = YourIAT_Trial_Data  %>%
  filter(grepl('Industrial|Innovative|Technical|Competitive|Tough|High-Performance|Powerful',trial_stimulus)) %>%
  group_by(trial_stimulus, study) %>% summarise_at(vars(reaction_time), list(average_reaction_time = mean))
feminine = YourIAT_Trial_Data %>%
  filter(grepl('Domestic|Aesthetic|Stylish|Helpful|Flexible|Colorful|Handcrafted',trial_stimulus)) %>%
  group_by(trial_stimulus, study) %>% summarise_at(vars(reaction_time), list(average_reaction_time = mean))

mrt <- dcast(masculine, study ~ trial_stimulus, value.var = "average_reaction_time")
mrt_cor <- cor(as.vector(unlist(mrt[1,2:ncol(mrt)])),as.vector(unlist(mrt[2,2:ncol(mrt)])), use = "complete.obs")

frt <- dcast(feminine, study ~ trial_stimulus, value.var = "average_reaction_time")
# correlation coeff for aes-domestic (feminine) terms
frt_cor <- cor(as.vector(unlist(frt[1,2:ncol(frt)])),as.vector(unlist(frt[2,2:ncol(frt)])), use="complete.obs")

cor.test(as.vector(unlist(mrt[1, 2:ncol(mrt)])),
       as.vector(unlist(mrt[2, 2:ncol(mrt)])),
       method = "pearson")

cor.test(as.vector(unlist(frt[1, 2:ncol(mrt)])),
       as.vector(unlist(frt[2, 2:ncol(mrt)])),
       method = "pearson")
```

Notes: